

SWAMI VIVEKANANDA, India

WMO: 428750

Lat: 21.183N

Lon: 81.733E

Elev: 318

StdP: 97.57

Time Zone: 5.50 (IND)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	12.0	13.3	0.4	4.0	27.2	2.5	4.7	26.7	6.0	23.1	5.1	21.2	0.7	0	0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.6	43.2	22.7	41.8	22.9	40.5	22.7	27.3	33.2	26.9	32.2	26.5	31.4	2.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	22.5	29.1	25.8	21.9	28.7	25.4	21.4	28.4	89.3	33.3	87.2	32.7	85.4	31.4	32.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.6	5.6	4.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	21.0	23.4	28.2	32.4	34.9	32.0	28.1	27.4	27.8	26.8	23.0	20.6		
	DBStd	4.83	1.99	2.12	2.60	2.17	2.57	3.36	1.85	1.44	1.22	1.63	1.85	1.75		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	7	4	0	0	0	0	0	0	0	0	0	0	3		
	CDD10.0	6259	341	377	564	673	772	661	560	539	534	522	389	329		
	CDD18.3	3224	87	143	306	423	513	411	302	281	284	263	139	73		
	CDH23.3	40272	665	1346	3922	6543	8561	6003	3292	2781	2998	2552	1047	563		
	CDH26.7	21326	121	520	2218	4288	6126	3750	1209	790	1039	919	258	87		
(14) Wind	WSAvg	1.7	0.8	1.3	1.4	2.1	1.9	2.8	2.5	2.3	1.8	1.3	1.1	1.2		
(15) Precipitation	PrecAvg	1237	11	13	18	13	21	174	368	334	218	56	6	7		
	PrecMax	1646	109	72	47	111	113	633	698	520	444	175	60	67		
	PrecMin	794	0	0	2	0	0	23	110	162	41	2	0	0		
	PrecStd	230	24	17	11	22	25	122	143	96	96	50	13	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	34.2	39.2	42.5	45.7	44.5	37.1	33.2	33.5	33.8	31.0	29.8		
		MCWB	16.9	18.4	20.8	21.2	21.9	25.3	24.9	26.2	24.9	23.5	20.5	18.7		
	2%	DB	29.0	32.8	37.9	41.2	44.2	41.9	34.6	32.0	32.7	32.8	30.0	28.7		
		MCWB	17.5	18.3	20.1	21.6	21.9	24.1	25.9	26.2	25.1	23.5	19.7	18.3		
	5%	DB	28.0	31.2	36.9	40.3	42.7	40.1	32.8	31.1	32.0	31.8	29.2	27.7		
		MCWB	17.5	18.3	20.2	21.3	22.6	24.4	26.2	26.0	25.2	23.7	19.3	17.9		
	10%	DB	27.0	30.0	35.7	39.2	41.4	38.2	31.5	30.2	31.2	30.9	28.5	26.7		
		MCWB	17.5	18.1	20.0	21.4	23.0	24.8	26.0	25.8	25.3	23.8	19.1	17.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.4	22.8	24.9	26.8	28.0	28.0	28.1	27.2	27.0	26.4	23.9	21.1		
		MCDB	27.2	30.3	35.3	37.6	41.0	36.6	32.3	30.1	30.6	29.9	26.7	27.1		
	2%	WB	20.4	21.6	22.8	25.1	26.6	27.2	27.3	26.8	26.6	25.9	22.7	20.1		
		MCDB	26.1	28.2	33.3	36.6	38.0	35.3	31.3	30.0	30.2	29.7	26.5	26.0		
	5%	WB	19.7	20.7	21.7	24.1	25.7	26.7	26.8	26.5	26.2	25.5	22.0	19.4		
		MCDB	25.3	26.8	33.5	35.9	36.8	34.1	30.6	29.6	29.7	29.2	26.3	25.4		
	10%	WB	19.0	19.8	20.9	23.1	24.9	26.2	26.4	26.2	26.0	25.2	21.1	18.5		
		MCDB	24.3	26.0	32.1	35.0	36.3	33.0	29.9	29.1	29.4	28.7	25.8	24.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	12.5	13.3	13.2	12.6	9.7	5.8	4.8	6.3	8.3	11.4	12.5		
		MCDBR	13.2	14.4	14.6	13.8	13.6	12.2	8.2	6.4	7.6	9.1	11.8	13.5		
	5% WB	MCWBR	5.0	5.0	5.1	4.4	4.2	3.1	2.6	2.0	2.4	2.9	3.8	5.2		
		MCWBR	10.2	10.8	13.3	12.8	12.5	10.2	6.7	5.7	6.5	7.3	8.5	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.465	0.475	0.536	0.555	0.600	0.656	0.594	0.555	0.549	0.528	0.513	0.472		
		taud	2.022	2.015	1.888	1.889	1.815	1.738	1.883	2.007	2.006	1.998	1.948	2.016		
	Ebn at Noon		794	816	783	773	732	686	731	763	762	758	741	768		
		Edh at Noon	160	170	201	203	217	233	201	178	176	171	171	156		
(44) All-Sky Solar Radiation	RadAvg		4.53	5.35	6.16	6.82	6.84	5.21	3.96	3.91	4.66	5.03	4.63	4.30		
	RadStd		0.24	0.23	0.30	0.25	0.16	0.53	0.47	0.37	0.44	0.40	0.26	0.22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page